PRODUCT HANDOFF · MOBILE APP

CallBridge mobile app: what it needs to do

Every feature, what each screen must show and do, and the states and edge cases to cover. There is no design language here on purpose: layout, navigation, visual style, motion, and iconography are the designer's call.

For the designer and mobile engineers

Works today in the web test client at callbridge.byte2bite.tech New for mobile not built yet

Where this says "screen", read "a place in the app": it can be a sheet, a tab, a step, or part of another screen. The server and its API stay the same; the mobile app is a new client for it.

The product in one paragraph

CallBridge makes phone calls for you in another language. You tell its assistant, in your own language, by voice or by typing, who to call and what you need: *"call my dentist and book a cleaning next week"*, *"tell Tabito I can't make Bible reading tonight"*, *"call the nearest Mexican restaurant and ask when they close"*. It works out the details, looks up anything it needs online, and checks the request against safety rules. You confirm, and an AI places the call in the other person's language, saying up front that it's an AI acting for you. You can follow the call live, read it translated, listen in, and answer the assistant's questions mid-call, or hand the call off completely. Afterwards you get the result in your language, and everything learned (the contact, the appointment, your decisions) is remembered for next time.

Who it's for: people who don't speak the local language comfortably. The first user is a Mandarin speaker in the US who also calls family and friends in Tagalog and Japanese.

Languages: the assistant talks with the user in their language (Mandarin, Cantonese, Spanish, Vietnamese, Korean, Tagalog, Russian, Arabic, Hindi, Japanese, English…) and calls in any of them. UI labels can stay English for v1, but everything generated (assistant speech, summaries, translations, rulings, questions) is in the user's language. Arabic means right-to-left text must render correctly wherever generated content appears.

Map of the app

1 Sign in	2 Profile interview
3 Home: start a call	4 Phone book
5 Review & confirm	6 Live call
7 Result	8 Talk it over
9 Call history New	10 Profile & privacy
11 Settings	12 Notifications New

Sign in Works today

The phone number is the account. No passwords, no email.

1. Enter a mobile number. US first: a fixed +1, formatted as you type, e.g. (901)-455-3148. Pasted or autofilled numbers that include +1 are accepted.
2. Choose the language "I'll speak". It's used for the profile interview and everything generated.

3. "Text me a code": a 6-digit code arrives by SMS. The code field supports the OS one-time-code autofill.
4. Enter the code. A new number creates the account and goes to the profile interview; a returning number goes home.

States and errors

- Sending… and checking…
- Wrong code: *"That code didn't work. Check it, or send a new one."* After 5 wrong tries the code is locked.
- Code expired after 10 minutes.
- Too many codes requested (limited per number and per network): try again in an hour.
- Invite-only: *"CallBridge is invite-only for now, and this number is not on the list."*
- Invalid number, or an emergency or premium-rate number.
- Actions: Send a new code · Change number.

A session lasts 90 days per device. Sign out and delete account live in Profile. Each code costs about $0.05, so the design shouldn't invite repeated sends.

Profile interview Works today

A short spoken or typed conversation with the assistant, in the user's language, that fills the profile so later calls need fewer questions. It follows the first sign-in and can be redone anytime ("Update by talking").

What it asks, one question at a time, each optional

1. What to call you: the name the AI uses on calls.
2. Pronouns, only if wanted.
3. Languages you speak, and which language calls are usually in.
4. Time zone or city, prefilled from the phone.

5. When you're usually free, e.g. weekdays after 5 pm.
6. Information the AI may share on calls when relevant: date of birth, home address, insurance provider name, an email. It's shared only when a call needs it, and confirmed per call.
7. Anything else: *"mornings are best"*, *"I'm a patient at Smile Dental"*.

Must support

- Talking and typing in the same conversation, switching freely (the same controls as Home).
- One-tap **Skip this question** and **That's all**, plus saying "skip" in any language.
- A live view of what's been saved, updating as the conversation goes.
- **Skip for now** and **Done**.
- Sensitive data is never stored: card numbers, bank details, SSNs, passwords, one-time codes. The assistant stops the user and explains; the server rejects them too, and the UI should show the item wasn't saved.

Home: start a call Works today

The main screen: one conversation with the assistant that ends in a call ready to confirm.

The conversation

- **Voice:** a mic control that starts the conversation, then turns the mic on and off without ending it. Mic permission is asked on first use.
- **Typing:** a text field into the same conversation. Typing never restarts anything; you can start by typing and never use the mic.
- The assistant speaks back and its words appear as text. A sound control mutes the voice but keeps the text.
- One-tap Skip this question and That's all. End the conversation without calling.

- Choosers: the language I'll speak, and the assistant's voice (10 voices, two marked most natural). The same voice is used on the call.

Statuses to communicate:

`connecting` `listening` `thinking`
`looking it up (~10 s quick, ~30–40 s thorough)` `speaking` `mic off`
`disconnected, with retry`

What the assistant gathers, in this order

1. **Who to call.** The number is found in this order and never asked if it can be found:
   a. the phone book, by name or by relationship in any language (*"my gf"*, *"女朋友"*, *"my dentist"*); it just confirms: *"Maria, your girlfriend, at 747-283-6440?"*
   b. online research for a kind of place or an unknown business (*"the nearest Mexican restaurant"*, *"a body shop that takes Geico"*);
   c. asking for the number, read back digit by digit.
2. **Why:** the purpose. For a personal message, the message itself.
3. A short questionnaire tailored to the kind of call, asking only what's missing, each question skippable. A clinic: who it's for, new or existing patient, reason, times, what may be shared. A restaurant: party size, times, name. A personal message: just confirm wording and language.
4. Call language: the contact's last language, otherwise the user's usual call language.

Live request card

Fills in as the conversation goes: who (name, relationship), number, the task in the user's language, call language, the ruling, and what's still missing.

Research results

- A short answer in the user's language, with sources.

- Up to 5 places to call: name, distance, open now, rating, why it fits (*"open now, takes Geico"*), address, number, and "in your phone book as …".
- An **unverified** marker on numbers that came from a web page; the assistant also reads those back. Tapping a place picks it.
- Location is asked the first time a "near me" question needs it. If denied, the assistant asks for a city or zip.

Same names

If more than one contact or place fits (two Marias, two locations of a chain), the assistant asks which one, by relationship, street, distance, or the last 4 digits. Make the options easy to tell apart.

Ground-rules ruling

Shown on the card, in the user's language:

- **Allowed:** appointments, reservations, service visits, questions to businesses.
- **Allowed with limits:** doctor, dentist, pharmacy (scheduling and status only); personal messages to people you know (delivered in your name; the AI ends the call if it's unwanted).
- **Not allowed**, with the reason: banking and payments; identity verification (insurance claims, government agencies, account changes); legal matters; emergencies (*"call 911 yourself"*); sales and marketing; anything deceptive, harassing, or pressuring.
- Missing details are listed: "Still needed: …".

Exits: Review & call (enabled once there's a number and a task; emphasized when the assistant says it's ready) and Use the form (manual entry, carrying over what was gathered). **Other ways in:** a contact's Call in the phone book (the assistant already knows who and only asks what for), and Talk it over from a result.

Phone book Works today

Everyone the user has called is saved automatically; the user can also add, edit, delete, and import.

Each contact: name; relationship (girlfriend, mom, dentist…); number; call language; address for businesses; kind (clinic, restaurant, personal…); notes learned on calls (*"asks for the insurance card at check-in"*); the last call's outcome in the user's language; number of calls.

Actions: Call (opens Home with the contact filled in) · Edit · Delete, with confirmation · Add · Search by name, relationship, or number.

Rules: contacts are unique by number, not by name, so two Marias can exist. A name the user sets is kept when later calls update the contact. A duplicate, invalid, emergency, or premium-rate number is refused with a reason.

Import from the phone's contacts New for mobile

- Use the OS contact picker so the user picks specific people. Only the picked names and numbers are sent to CallBridge; there's no whole-address-book upload.
- After picking, let the user set relationship and call language (both optional), and choose a number when a contact has several.
- If the number is already in the phone book, offer to update it instead of adding a duplicate.

Review & confirm Works today

Shown before every call, whether it came from the conversation or the form. Everything is editable:

- who and number; the task in the user's language; call language;
- when the user is available (time windows; required for calls that book a time);
- extra charges the AI may accept (default none);

- what it may share on this call, picked from the profile's shareable information or added here;
- the user's name and pronouns as the AI will use them; voice.

During the call: a choice before every call

The last choice is remembered.

- **Stay in the loop:** if the other side asks for something outside the limits, the AI puts them on hold and asks the user in the app, for up to 60 seconds. The user can also message the AI mid-call.
- **Hand it off completely:** no interruptions. The AI works within the limits, politely declines anything else (*"Leo will follow up"*), and the user just gets the result.

Consent: the user confirms they're asking for this call themselves, to a specific business or someone they know, and that the AI will say it's an AI. Start call is disabled until then.

Errors: missing or invalid fields jump to what needs fixing; refused by the ground rules (reason in the user's language); another call already in progress (one at a time); hourly limit reached.

Live call Works today

Progress, with a call timer:

`preparing` `dialing` `ringing` `connected` `in progress` `wrapping up` `done`

Failures: no answer, busy, voicemail (the AI leaves no personal details), failed with a reason.

Transcript

- Both sides, plus the user's own messages to the AI, marked as not heard by the other party.

- Each line in the user's language first, with the original underneath. Translations arrive about 1–2 seconds after each line; if the call is in the user's language there's none.
- Lines the AI was interrupted on are marked.

Listen live

A control to hear both sides in real voice, about a quarter second behind. Listen-only. It starts from a tap (OS audio rules) and stops when the call ends.

Questions on hold (Stay in the loop only)

When the AI needs a decision, a prominent, urgent question appears:

- the question in the user's language with the original underneath, plus the amount, date, or time when relevant;
- a countdown, 60 seconds by default, after which the AI says the user will follow up and moves on;
- **Approve / Decline / Reply…** For questions like "we need your insurance info", Reply comes first and Decline reads "Don't share";
- a typed reply is blocked if it contains card, SSN, password, or account numbers, with the reason;
- several can stack; answered ones collapse into a history line ("You approved…", "No answer in time…");
- it must get attention when the app is in the background (see Notifications).

Message the assistant (Stay in the loop only)

A text field to steer mid-call: *"tell them I'll be 10 minutes late"*. It can't widen what the AI may agree to; anything outside the limits comes back as a question. In Hand it off mode, show a calm note instead: the assistant won't interrupt, and the result comes at the end.

End call hangs up. **Leave** keeps the call going; the user can return from history or the notification.

Handled by the AI, visible in the transcript: call screening (it says who it is and, for personal calls, the message itself, then waits), hold music, phone menus (it can speak options but not press keys), the other side hanging up.

Result Works today

Shown when the call finishes, and from history and the "call finished" notification.

- **Headline** in the user's language: *"Booked: Thu Oct 8, 2:00 pm at Smile Dental"* or *"Not booked: they only had 3 pm"*.
- **Appointment** if one was made: day, date, time, what. If the other side may not have agreed, it's labeled **needs confirmation**, not confirmed.
- **Summary** in the user's language, with English underneath when different.
- **You decided during the call:** approved, declined, answered.
- **Needs your answer:** questions left open.
- **The assistant declined:** things it refused on the user's behalf.
- **Next steps** in the user's language.
- **Please double-check:** warnings, such as something said that the rules didn't validate.
- Facts: extra charges authorized or not; number of commitments.
- Transcript, expandable and translated like the live view.
- Actions: Talk it over · New call · Add to calendar New for mobile.

Talk it over Works today

A voice or typed conversation about a finished call. It opens by telling the user how it went, in their language. It answers questions from the call record and transcript only (*"did they mention insurance?"* → *"it didn't come up"*), and can set up a follow-up call from the previous one (*"call back and take 3 pm"*), asking only what changes. Same controls as Home.

Call history New for mobile

The user's calls, newest first: who, when, status, and the headline. Live calls are marked and open the live screen; finished ones open the result. The API for this already exists.

Upcoming appointments from the profile belong here or on Home: date, time, with whom, and "needs confirmation" when flagged.

Profile & privacy Works today

- Header: name, phone number, languages, time zone.
- Update by talking, which reopens the interview.
- Sections, each item deletable: usually free · may be shared on calls · preferences · upcoming appointments · contacts (linking to the phone book) · decisions from calls.
- Sign out. Delete account, with confirmation: removes the profile, phone book, and call history.
- Never stored: card numbers, bank details, SSNs, passwords, one-time codes. Attempts are refused with a reason.

Settings

Voice · language I speak · default "during the call" choice · notification preferences · legal and about · sign out. The web client's server URL and demo mode are developer-only and stay out of the production app. A sample call for first-time users is a nice-to-have.

Notifications New for mobile

- **Question on hold:** high priority, with Approve and Decline actions on the notification itself (Reply opens the app), and the deadline shown. This one

matters most: the other party is waiting.

- **Call finished**, with the headline: *"Booked Thu 2 pm at Smile Dental"*.
- **Call failed, no answer, or busy.**

Permissions, asked in context

PERMISSION	WHEN	IF DENIED
Microphone	First tap on the mic	Typing still works
Location	First "near me" research	The assistant asks for a city or zip
Contacts	First "Import from contacts"	Add contacts manually
Notifications	Before the first call, explaining hold questions and results	In-app alerts only; hold questions may time out

Content and tone

- The assistant is warm and brief. Everything generated is in the user's language.
- The other party is always told it's an AI. The app should never imply the user is on the call.
- Calls cost a few cents a minute; lookups and sign-in codes a few cents each. There's no pricing UI in v1, but don't design for repeated retries.

Not in v1

Ideas, not requirements:

- **Join the call:** the user taps in and speaks to the other party directly, taking over from the AI.
- Answering hold questions by voice.

- Calling from the user's own number as caller ID.
- Scheduling a call for later; recurring calls.

For engineers

- **API:** the same HTTP API as the web client: `/api/auth/*`, `/api/me*`, `/api/intake/*`, `/api/research`, `/api/calls*`, and the `/listen/:id` WebSocket. The server is the source of truth for every rule; keep the client thin.
- **Auth:** `Authorization: Bearer <session token>` from `/api/auth/verify`, stored in the Keychain or Keystore.
- **Voice conversation:** OpenAI Realtime over WebRTC with a 60-second key from `/api/intake/session`, via `react-native-webrtc` (so a custom dev build, not Expo Go). The client relays the tool calls (`update_request`, `check_request`, `research`, `finish_intake`, `update_profile`, `finish_profile`) the way `web/src/voiceIntake.ts` does.
- **Live call:** poll `GET /api/calls/:id` about once a second, or use the `/api/calls/:id/stream` server-sent events; answer questions with `POST /api/calls/:id/questions/:qid`; send messages with `POST /api/calls/:id/messages`; hang up with `POST /api/calls/:id/hangup`. To listen, get a ticket from `POST /api/calls/:id/listen`, then open `/listen/:id?ticket=…` (base64 μ-law 8 kHz frames tagged `them` or `ai`, plus `clear` and `end`).
- **Push** is new server work: register device tokens, and send on a question on hold, call finished, and call failed.
- **Contacts import:** `POST /api/me/contacts` for each picked contact.

The same document lives in the repo as `docs/MOBILE_APP_HANDOFF.md`.